

Competitive Audio-Language Models with Data-Efficient Single-Stage Training on Public Data

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Abstract—Large language models (LLMs) have transformed NLP, yet their integration with audio remains underexplored—despite audio’s centrality to human communication. We introduce Falcon3-Audio, a family of Audio-Language Models (ALMs) built on instruction-tuned LLMs and Whisper encoders. Using a remarkably small amount of public audio data—less than 30K hours (5K unique)—Falcon3-Audio-7B matches the best reported performance among open-weight models on the MMAU benchmark, with a score of 64.14, matching R1-AQA, while distinguishing itself through superior data and parameter efficiency, single-stage training, and transparency. Notably, our smallest 1B model remains competitive with larger open models ranging from 2B to 13B parameters. Through extensive ablations, we find that common complexities—such as curriculum learning, multiple audio encoders, and intricate cross-attention connectors—are not required for strong performance, even compared to models trained on over 500K hours of data.

Index Terms—audio-language models, speech understanding, music understanding, sound understanding, data-efficient training, large language models

I. INTRODUCTION

Large language models (LLMs) like GPT-4 [1], Gemini 1.5 [2], LLaMA 3 [3], Falcon 3 [4], Qwen2 [5], and Phi-3 [6] have demonstrated advanced capabilities in text understanding, reasoning, and generation. With enhancements in instruction tuning and dataset curation techniques [7]–[9], their capabilities continue to grow. However, the next frontier is multimodal intelligence, extending these powerful models to rich modalities like audio.

Audio is central to human communication, conveying not only linguistic content but also paralinguistic information like tone, emotion, and speaker intent. Enabling LLMs to understand audio could unlock powerful applications in assistive AI, conversational agents, sound event detection, and music reasoning. However, audio-language modeling remains underexplored compared to vision-language modeling, and presents unique challenges: (1) lack of established design principles for audio-language models (ALMs), (2) high diversity across audio domains (speech, music, environmental sounds), (3) high temporal resolution of audio signals, and (4) variable input lengths that complicate alignment with LLM input sequences.

Many ALMs try to solve such challenges with increasingly complex designs—multi-stage pipelines, large proprietary datasets, or intricate cross-modal attention mechanisms. This diversity reflects a lack of consensus and creates barriers to reproducibility and community-driven progress. There is a

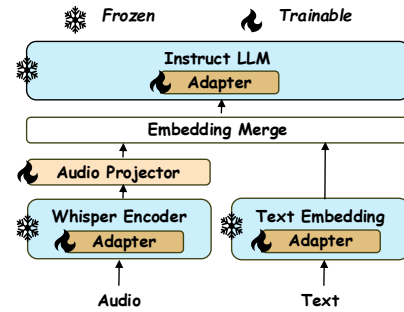


Fig. 1: Architecture of Falcon3-Audio which integrates Whisper audio encoder to extract features, and projects audio tokens into the Instruct LLM input space via a learnable projector.

growing need for ALMs that are not only competitive but also transparent, efficient, and fully based on public resources.

Our Contribution: We present **Falcon3-Audio**, a family of ALMs that offer a simple, transparent, and reproducible architecture for aligning audio with instruction-tuned LLMs. Falcon3-Audio is built entirely on public data, with *models of 1B, 3B, and 7B parameters and trained end-to-end via a single-stage fine-tuning strategy*. Through a systematic exploration of critical design choices, *aiming for a minimalist approach*, we show that strong audio-language performance does not require complex architectures, proprietary data, or multi-stage training.

- **Efficient Architecture:** Falcon3-Audio connects Whisper audio encoder to an instruction-tuned LLM through a lightweight projection module.
- **Strong Results with Public Data:** Falcon3-Audio-7B matches state-of-the-art performance among open-weight models on the MMAU benchmark, despite using less than 30K hours of audio - significantly less than similar performing models. It also achieves strong performance on AIR-Bench Foundational and Chat.
- **Transparent and Reproducible Design:** Falcon3-Audio uses only public datasets and open-source models, making it easy to reproduce, extend, and evaluate.
- **Comprehensive Ablation Studies:** We conduct detailed ablations across design factors—audio encoder, sequence length, projection design, training strategy, and data mixture—providing practical guidance for future ALMs.

II. RELATED WORK

Building upon the remarkable progress in LLMs, our work on Falcon3-Audio is deeply informed by advancements in both vision-language models and the rapidly evolving audio-language models.

A. Vision Language Models: Insights for Multimodal Integration

Vision-language models (VLMs) have pioneered the integration of non-textual modalities into LLMs, achieving impressive results in tasks such as visual question answering, image captioning, and multimodal reasoning. Models like Flamingo [10], LLaVA [11], ShareGPT4V [12], Qwen-Vision [13], and IDEFICS [14] demonstrate effective strategies for combining image and text inputs. These models commonly leverage powerful pre-trained unimodal encoders—such as CLIP [15] or SigLIP [16] for vision, and LLaMA or Mistral for language—integrating them through cross-attention or projection-based mechanisms. VLMs often interleave image features within the LLM architecture using cross-attention, or directly inject visual embeddings by concatenating patch-wise image representations with text tokens.

Crucially, the VLM field offers valuable architectural patterns and integration techniques applicable to other modalities. For instance, the concept of treating frame-wise spectrogram features from audio encoders (like Whisper) as analogous to patch-wise image embeddings in VLMs provides a foundational parallel for Audio-Language Model design. While both VLMs and ALMs face challenges in feature alignment and scaling, audio introduces unique complexities such as variable input lengths and the diversity of audio domains (speech, music, environmental sounds). The progress in VLMs, particularly in efficient feature integration and projection, serves as a crucial stepping stone and inspiration for extending LLMs to the audio modality, informing our approach in Falcon3-Audio.

B. Audio Language Models: Towards General Audio Understanding

The field of Audio-Language Models (ALMs) is rapidly emerging as a critical area of research, aiming to equip LLMs with the ability to understand and reason with audio. Early ALM research focused on cross-modal encoders like AudioCLIP [17] and CLAP [18], primarily for tasks like audio retrieval and classification by aligning audio and text representations in a shared embedding space. The advent of powerful LLMs has shifted the paradigm towards directly integrating audio processing within LLMs, unlocking more sophisticated audio-grounded reasoning and generation capabilities. This has led to specialized ALMs like MuLLama [19] and MusiLingo [20] for music understanding, and LTU [21] for non-speech audio tasks. SLAM-ASR [22] is a speech recognition-focused effort, complementary to our work. Recent advances in acoustic tokenization, such as WavTokenizer [23], have enabled efficient compression of audio signals into discrete tokens, facilitating scalable speech generation with large language model backbones. These early LLM-based

ALMs demonstrated the potential of audio-text alignment but often remained limited in scope and generalization.

More recent advancements have focused on developing general-purpose ALMs capable of handling a wider range of audio tasks. Models like Pengi [24], LTU-AS [25], SALMONN [26], GAMA [27], Qwen-Audio [28], and Qwen2-Audio [29] represent this trend, integrating diverse audio types (speech, environmental sounds, music) into LLM frameworks. These models use various techniques: LTU-AS replaced AST encoders with Whisper and incorporated time/layer transformers; SALMONN uses cross-attention to fuse features from multiple encoders like Whisper and BEATs [30]; GAMA utilizes multi-layer transformer aggregators; and the Qwen-Audio series has progressively refined training frameworks and data scale. Qwen2-Audio, in particular, streamlined training with natural language prompts and expanded to multi-turn audio interactions.

Concurrent Works: Several concurrent works—such as R1-AQA [31], Audio Flamingo 2 [32], Phi-4 Mini [33], and Qwen2.5 Omni [34]—were released after the completion of our experiments. While relevant, they are excluded from some of our evaluation tables due to incomplete reporting: none provide results on AIR-Bench Foundational [35], and only Phi-4 Mini reports on AIR-Bench Chat [35]. For MMAU [36], Qwen2.5 Omni reports only on the test-mini split. **R1-AQA** fine-tunes Qwen2-Audio Instruct using Group Relative Policy Optimization (GRPO) on 38K task-aligned samples. It inherits Qwen2-Audio Instruct’s dependency on >500K hours of non-public data, a large 8.4B model, and a multi-stage training pipeline—with GRPO introducing an additional fine-tuning stage. **Audio Flamingo 2**, while transparent, uses a more complex setup: a separately pretrained CLAP [18] encoder is fused into a 3B LLM via cross-attention layers within intermediate transformer blocks. This requires careful finetuning to preserve the LLM’s language capabilities. Falcon3-Audio, in contrast, employs a simple projection into the LLM’s input embeddings and trains end-to-end in a single stage. **Phi-4 Mini** is a 5.6B model trained on over 2 million hours of private speech data, in addition to non-speech audio, images, videos, and text. It uses a multi-stage curriculum with vision and audio adapters, but provides limited transparency in terms of training data sources and fine-tuning details. **Qwen2.5 Omni** also incorporates a vision encoder, like Phi-4 Mini, but uses a larger 10.7B model. It is trained on considerably more data than Falcon3-Audio—sourced from undisclosed datasets—and follows a multi-stage curriculum. The lack of transparency in both data and training pipeline limits reproducibility.

Despite these recent advancements, many of these models rely on complex architectures and large-scale proprietary data. Falcon3-Audio, by contrast, achieves strong performance using a simpler, transparent setup based solely on public resources. Full performance comparisons are presented in Section V-B.

III. DESIGN CHOICES: GUIDING PRINCIPLES FOR FALCON3-AUDIO

To realize Falcon3-Audio as an efficient, performant, and accessible audio-language model, we conducted a systematic exploration of key design elements. This section details the critical considerations that shaped our final model. Our goal was to identify a configuration that optimally balances simplicity, computational efficiency, and state-of-the-art performance. Detailed results of our ablation studies are presented in the supplementary material (Section I).

A. Language Model Foundation: Base or Instruct-Tuned?

For Falcon3-Audio’s language backbone, we chose a Falcon3 [4] family of instruction-tuned LLMs, available in 1B, 3B, and 7B variants, which are established strong performers within their respective size categories. For the 7B variant, we used an internal version trained with less code and math data compared to the released model, as we observed improved performance with this configuration. Our decision to utilize instruction-tuned models, rather than base model counterparts, was driven by the hypothesis that their inherent instruction-following capabilities would be advantageous for audio-language tasks, often involving complex instructions and queries. Empirical validation confirmed this hypothesis.

B. Audio Encoder Selection: Capturing the Spectrum of Audio Information

Selecting an appropriate audio encoder is crucial for extracting audio features that effectively generalize across the diverse landscape of audio tasks. While options like AST [37], CLAP [18], ImageBind Audio Encoder [38], and BEATs [30] each offer unique strengths, we prioritized Whisper [39] for Falcon3-Audio. Whisper has emerged as a leading encoder, particularly for speech-related tasks, and has demonstrated strong performance even in non-speech audio domains [29], [40]. Given the importance of speech understanding and Whisper’s overall robustness, we used the Whisper Medium English encoder for our 7B/3B models, and the Whisper Small English encoder for 1B, balancing performance and parameter efficiency. Ablation studies confirmed that Whisper Medium (English) provided the best performance for our 7B model.

C. Audio Sequence Handling: Balancing Efficiency and Temporal Resolution

Audio signals, characterized by high temporal density, pose a challenge for efficient processing. Even Whisper’s down-sampling to 50 tokens per second results in a much higher token rate compared to text. Approaches to audio sequence reduction vary. LTU-AS [25] uses time and layer transformers for aggressive reduction. Shuka-v1 [41] utilizes stacking from SLAM-ASR [22] to reduce sequence length. Based on ablation studies, we found that aggressive sequence length reduction was not necessary for our architecture and tasks. Therefore, we adopted a straightforward pooling layer, similar to Qwen2-Audio [29], which reduces the token rate to 25 tokens per second. This provides a computationally efficient solution.

D. Connector Architecture: Bridging Audio and Text Embeddings

The architecture connecting the audio encoder and the language model is a critical design choice. We followed a projection-based approach, proven successful in VLMs like LLaVA-v1.5 [42] and ALMs like Qwen2-Audio [29]. Falcon3-Audio uses a learnable projector to map the audio encoder’s output dimensionality to the LLM’s input dimension. This projector is augmented with an activation function, an additional linear layer, and layer normalization, inspired by Shuka-v1 [41]. Ablation studies showed that incorporating intermediate layer features from the Whisper encoder did not improve performance, so we opted for the simpler final-layer projection.

E. Training Data Composition: Ensuring Generalization Across Audio Domains

The selection and composition of training data are paramount for developing ALMs capable of generalizing to a wide range of audio-related tasks. For Falcon3-Audio, we adopted the Open-ASQA dataset from LTU-AS [25] as our primary training corpus. Open-ASQA, the largest publicly available audio instruction dataset at the time of our development, comprises approximately 10 million samples (26K hours) drawn from diverse sources, including AudioSet [43], AudioCaps [44], FSD50K [45], FreeSound [46], LibriTTS [47], VGGSound [48], and VoxCeleb [49]. To further enhance voice interaction capabilities, we supplemented this with 440K samples (approximately 1K hours) from a synthetic voice-instruction dataset [50]. Ablation studies indicated that adding further datasets did not improve performance on our target benchmark.

F. Training Strategy: Simplicity vs. Curriculum Learning

In contrast to complex, multi-staged curriculum learning approaches, we investigated a simpler *single-stage end-to-end fine-tuning* strategy. This strategy trains the entire model on the full task spectrum from the outset. Ablation studies showed that this single-stage approach was superior to a two-stage curriculum for our model. Furthermore, we found that training all modules jointly (audio encoder, projector, and LLM) yielded the best results.

IV. METHODOLOGY

Falcon3-Audio integrates a pretrained Whisper audio encoder, a projection module, and an Instruct-tuned LLM for efficient audio-language understanding (Figure 1). We describe each component and our single-stage end-to-end training strategy below.

A. Falcon3-Audio Architecture

1) *Audio Encoder*: Falcon3-Audio uses the Whisper model [39] to extract audio features. We use Whisper Medium (English) for the 7B and 3B variants and Whisper Small (English) for the 1B variant.

TABLE I: MMAU Benchmark (Test) results. TDTS: Transparent Data and Training Strategy; Train Data: audio hours. **Top scores** are bolded; second-best are underlined; values are left empty where unavailable; *indicates concurrent work. Sources (except Falcon3-Audio): [33], [36].

Rank	Method	TDTS	Model Size	Train Data	Average	Sound	Music	Speech
<i>End-To-End ALMs</i>								
1	R1-AQA ⁺	✗	8.4B	500K-2M	64.36	69.76	61.40	62.70
2	Falcon3-Audio 7B	✓	7.8B	0-30K	<u>64.14</u>	71.27	<u>58.53</u>	<u>62.63</u>
3	Audio Flamingo 2 ⁺	✓	4.7B	30K-100K	59.42	65.10	72.90	40.26
4	Falcon3-Audio 3B	✓	3.6B	0-30K	57.96	67.33	52.07	54.47
5	Phi-4-Mini ⁺	✗	5.6B	2M+	55.56			
6	Qwen2-Audio Instruct	✗	8.4B	500K-2M	52.50	45.90	53.26	45.90
7	Falcon3-Audio 1B	✓	1.8B	0-30K	50.80	59.90	46.30	46.20
8	Qwen-Audio Chat	✗	8.4B	100K-500K	41.86	56.73	40.90	27.95
9	SALMONN	✓	13B	0-30K	32.77	40.30	33.76	24.24
10	GAMA	✓	7B		31.81	45.40	30.83	19.21
11	MuLLaMA	✓	7B		30.66	44.80	30.63	16.56
12	GAMA-IT	✓	7B		29.02	43.23	28.00	15.84
13	LTU-AS	✓	7B	0-30K	18.90	24.96	10.46	21.30
14	Audio Flamingo Chat	✓	2.2B	0-30K	18.87	28.26	18.20	10.16
15	LTU	✓	7B	0-30K	18.51	25.86	12.83	16.37
16	MusiLingo	✓	7B		13.39	27.76	6.00	6.42
17	M2UGen	✓	7B		12.87	3.69	30.40	4.53
<i>Baselines</i>								
	Human (Test-Mini)				82.23	86.31	78.22	82.17
	Random Guess				25.92	25.73	26.53	25.5
	Most Frequent Choice				26.5	25.73	23.73	30.33
<i>APIs And Cascaded Approaches</i>								
	Qwen2-Audio Instruct + GPT4o				58.74	55.83	51.73	68.66
	Qwen2-Audio Instruct + Llama-3-Instruct				53.57	49.10	62.70	55.25
	Gemini Pro v1.5				52.97	54.46	48.56	55.90
	EnCLAP/MuLLaMA/Whisper + GPT4o				48.65	35.80	39.52	68.27
	EnCLAP/MuLLaMA/Whisper + Llama-3-Instruct				45.87	33.73	42.36	61.54

Given an audio sequence x of length T , the Whisper encoder produces a feature representation h_{audio} through a series of L layers:

$$h^{(l)} = f_{\text{AE}}(h^{(l-1)}; \theta_{\text{AE}}^{(l)}), \quad h^{(0)} = x \quad (1)$$

$$h_{\text{audio}} = h^{(L)} \quad (2)$$

where $h^{(l)}$ is the feature at layer l , f_{AE} is the transformation, and $\theta_{\text{AE}}^{(l)}$ are the parameters.

2) *Projection Module*: A lightweight projection module aligns h_{audio} with the LLM's embedding space. It consists of: an initial layer normalization, a linear layer (W_1, b_1), GELU activation (σ), a second linear layer (W_2, b_2), and a final layer normalization:

$$h_{\text{norm1}} = \text{LayerNorm}(h_{\text{audio}}) \quad (3)$$

$$h_{\text{proj1}} = W_1 h_{\text{norm1}} + b_1 \quad (4)$$

$$h_{\text{act}} = \sigma(h_{\text{proj1}}) \quad (5)$$

$$h_{\text{proj2}} = W_2 h_{\text{act}} + b_2 \quad (6)$$

$$h_{\text{final}} = \text{LayerNorm}(h_{\text{proj2}}) \quad (7)$$

3) *Large Language Model*: Falcon3-Audio uses Falcon3 Instruct LLMs as its core. Given a tokenized text input z with embedding $\text{EMB}(z)$, Falcon3-Audio processes the merged sequence:

$$h_{\text{input}} = [h_{\text{final}}, \text{EMB}(z)] \quad (8)$$

The LLM then generates a probability distribution over text tokens:

$$p(y|x, z) = f_{\text{LLM}}(h_{\text{input}}; \theta_{\text{LLM}}) \quad (9)$$

where f_{LLM} is the LLM and θ_{LLM} are its parameters.

B. Single-Stage End-to-End Training

Falcon3-Audio is trained end-to-end, maximizing the likelihood of generating correct completion tokens. The training dataset, $\mathcal{D}_{\text{Train}}$, combines Open-ASQA [25] (approximately 10 million samples, 26K hours) and a synthetic voice instruction dataset (440K samples, 1K hours):

$$\mathcal{D}_{\text{Train}} = \mathcal{D}_{\text{Open-ASQA}} \cup \mathcal{D}_{\text{Voice-Instruction}}$$

We fine-tune the projection module, audio encoder, and LLM using Low-Rank Adaptation (LoRA). The training objective is the negative log-likelihood:

$$- \sum_{(x, z, y) \in \mathcal{D}_{\text{Train}}} \log p(y|x, z; \theta_{\text{Projector}}, \theta_{\text{AE}}^{\text{LoRA}}, \theta_{\text{LLM}}^{\text{LoRA}})$$

where $\theta_{\text{Projector}}$, $\theta_{\text{AE}}^{\text{LoRA}}$, and $\theta_{\text{LLM}}^{\text{LoRA}}$ are the parameters of the projection module and the LoRA-adapted parameters of the audio encoder and LLM, respectively. This approach ensures robust audio-text alignment and efficient use of public data.

TABLE II: AIR-Bench Foundational Benchmark results. **Top scores** are bolded; second-best are underlined.

Task	Falcon3-Audio 7B	Qwen2-Audio Instruct	Falcon3-Audio 3B	Falcon3-Audio 1B
Rank	1	2	3	4
Total Average	54.00	<u>44.00</u>	42.00	38.00
Sound Average	65.2	49.8	<u>53.4</u>	47.7
Audio grounding	70.70	17.80	<u>60.00</u>	42.10
Vocal sound classification	90.70	71.10	<u>74.90</u>	72.10
Acoustic scene classification	43.60	40.50	<u>40.80</u>	35.20
Sound question answering	71.60	<u>62.80</u>	52.40	50.40
Music Average	50.5	<u>46.1</u>	42.2	36.4
Music instruments classification	58.50	<u>49.60</u>	46.90	39.50
Music genre classification	65.10	<u>63.90</u>	59.90	50.60
Music note analysis-pitch	28.80	<u>24.30</u>	19.60	23.90
Music note analysis-velocity	25.00	<u>24.70</u>	22.80	19.10
Music question answering	58.70	<u>56.00</u>	41.80	38.80
Music emotion detection	45.70	<u>38.70</u>	<u>39.90</u>	29.50
Speech Average	50.4	<u>43.5</u>	35.1	34.2
Speech grounding	22.20	<u>26.30</u>	20.30	26.90
Spoken language identification	66.00	<u>38.10</u>	4.90	13.70
Speaker gender recognition	58.50	<u>52.50</u>	23.90	29.60
Emotion recognition	64.10	35.40	<u>60.30</u>	52.60
Speaker age prediction	<u>24.20</u>	22.30	26.40	8.30
Speech entity recognition	36.90	48.30	26.20	<u>46.90</u>
Intent classification	<u>71.70</u>	78.00	51.40	44.70
Speaker number verification	<u>38.30</u>	38.00	39.20	21.10
Synthesized voice detection	<u>49.80</u>	51.90	49.30	49.70

TABLE III: AIR-Bench Chat Benchmark results. Train Data: audio hours . **Top scores** are bolded; second-best are underlined; values are left empty where unavailable; +indicates concurrent work. Sources (except Falcon3-Audio): [30], [33].

Rank	Model	Model Size	Train Data	Average	Speech	Sound	Music	Mixed
<i>End-To-End ALMs</i>								
1	Phi-4-Mini+	5.6B	2M+	6.98				
2	Qwen2-Audio Instruct	8.4B	500K-2M	<u>6.93</u>	7.18	6.99	6.79	6.77
3	Falcon3-Audio 7B	7.8B	0-30K	6.20	<u>6.93</u>	6.74	5.64	5.48
4	SALMONN	13B	0-30K	6.11	<u>6.16</u>	6.28	<u>5.95</u>	<u>6.08</u>
5	Qwen-Audio Chat	8.4B	100K-500K	6.08	6.47	<u>6.95</u>	5.52	5.38
6	Falcon3-Audio 3B	3.6B	0-30K	5.88	6.65	<u>6.34</u>	5.37	5.17
7	Falcon3-Audio 1B	1.8B	0-30K	5.67	6.33	6.14	5.48	4.74
8	BLSP	7B		5.33	6.17	5.55	5.08	4.52
9	Pandagpt	7B		4.25	3.58	5.46	5.06	2.93
10	Next-gpt	8.2B		4.13	3.86	4.76	4.18	2.92
11	SpeechGPT	13B		1.15	1.57	0.95	0.95	1.14
12	Macaw-LLM	7B		1.01	0.97	1.01	0.91	1.00
<i>APIs</i>								
	Qwen-Audio-Turbo			6.34	7.04	6.59	5.98	5.77
	Gemini-1.5-pro			5.70	6.97	5.49	5.06	5.27

V. EXPERIMENTAL EVALUATION

A. Experimental Setup

Model Variants: To evaluate scalability and deployment flexibility, we trained Falcon3-Audio in three model sizes: 7B, 3B, and 1B parameters. Each variant shares a consistent architecture. Table IV provides a detailed parameter breakdown for each Falcon3-Audio model size.

Training Infrastructure: Our training pipeline was built using the Qwen2AudioForConditionalGeneration class from *HuggingFace Transformers*. We executed distributed training via *HuggingFace Accelerate* with DeepSpeed ZeRO-2 op-

timization and the *HuggingFace TRL*'s SFT Trainer. Total training time was approximately 20 hours.

Compute Resources: Falcon3-Audio models were trained on AWS SageMaker, utilizing 8 nodes of P5 instance types, each equipped with 8×80GB H100 GPUs.

Hyperparameter Settings and the **Prompt Format** are shown in the supplementary material (Section II).

B. Performance Evaluation

We evaluated Falcon3-Audio on the MMAU benchmark (10K samples) [36] for multi-domain audio understanding,

TABLE IV: Parameter breakdown of Falcon3-Audio models.

Variant	AE	Proj.	LLM	Total
7B	307M	12M	7.4B	7.8B
3B	307M	12M	3.2B	3.6B
1B	88M	5M	1.7B	1.8B

and the AIR-Bench Foundational (22K samples) and AIR-Bench Chat (2.2K samples) benchmarks [35] for task-specific and open-ended audio understanding, respectively. See the supplementary material (Section III) for more details on these benchmarks. We fixed all random seeds (Python, NumPy, PyTorch, CUDA) to 0. On both MMAU and AIR-Bench Foundational, with a non-zero temperature, this yielded identical discrete answer strings, out of four choices per sample, across three runs.

1) *MMAU - Matching State-of-the-Art with Data Efficiency:* For MMAU evaluation, we configured Falcon3-Audio by setting temperature = 0.1, top-p = 0.8, top-k = 500, and a maximum of 32 new tokens. Table I summarizes results across related models.

Falcon3-Audio-7B achieves an average accuracy of 64.14% on the MMAU test set — matching the top-performing R1-AQA, which fine-tunes Qwen2-Audio Instruct using GRPO on 38K samples. In contrast, Falcon3-Audio is trained via a significantly simpler single-stage strategy, using less than 30K hours of public audio, without any curriculum learning or GRPO-style policy optimization. Qwen2-Audio Instruct, by comparison, uses over 500K hours of undisclosed data, and R1-AQA adds an additional GRPO stage on top. Falcon3-Audio also has a smaller parameter footprint than Qwen2-Audio Instruct (8.4B), R1-AQA (8.4B), and SALMONN (13B).

Despite over 90% overlap in training data with LTU-AS, Falcon3-Audio models demonstrate a substantial performance gain over LTU-AS’s 18.90% average accuracy. Notably, even our smallest variant, Falcon3-Audio-1B, ranks just behind Qwen2-Audio Instruct and outperforms many larger open-weight models in the 2B–13B range. These results show that strong generalization across diverse audio domains is achievable without massive data or complex multi-stage pipelines.

2) *AIR-Bench Foundational - Clear Gains in Instructional Audio Tasks:* We evaluated Falcon3-Audio on the AIR-Bench Foundational benchmark, which covers a broad set of instruction-following tasks across speech, sound, and music. To ensure fair comparison, we corrected limitations in the original scoring script, which (1) ignored invalid outputs (e.g., empty or malformed generations), and (2) checked only for answer option identifiers (e.g., “A”, “B”) without verifying the actual answer content. Our revised scorer treats invalid generations as incorrect and matches the answer content textually (see supplementary material, Section IV), similar to the MMAU evaluation.

Due to the compute budget, we compare Falcon3-Audio primarily with Qwen2-Audio Instruct, a strong baseline on both MMAU and AIR-Bench Chat. As shown in Table II,

Falcon3-Audio-7B outperforms Qwen2-Audio Instruct by a wide margin of over 10 percentage points. Falcon3-Audio-3B also performs competitively, narrowing the gap significantly. Importantly, Falcon3-Audio was trained on a relatively modest (<30K hours), general-purpose dataset without tailoring for specific AIR-Bench tasks. In contrast, Qwen2-Audio Instruct benefits from multi-stage training and over 15× more data from undisclosed sources. These results highlight Falcon3-Audio’s strong instruction-following capabilities and broad generalization.

3) *AIR-Bench Chat - Competitive Results:* We evaluated Falcon3-Audio on the AIR-Bench Chat using GPT-4 Turbo, following the official setup (temperature 1.0 with a 50% chance of switching to 2.0). We ran three evaluation trials and report the median score. Falcon3-Audio used the same generation settings as in Section V-B1, except with a repetition penalty of 1.1 and a maximum of 512 new tokens.

As shown in Table III, Falcon3-Audio-7B ranks third among end-to-end open-weight ALMs, behind Phi-4 Mini and Qwen2-Audio Instruct. Both rely on significantly larger and less transparent pipelines: Phi-4 Mini is trained on over 2M hours of multimodal data with a multi-stage curriculum, while Qwen2-Audio Instruct uses over 500K hours of data from undisclosed sources. In contrast, Falcon3-Audio is trained on <30K hours of public audio in a single-stage setup, yet remains competitive.

It is worth noting that Falcon3-Audio-7B outperforms both Phi-4 Mini and Qwen2-Audio Instruct by around 10 points on MMAU (Section V-B1). Falcon3-Audio-7B also surpasses Qwen2-Audio Instruct on AIR-Bench Foundational (Section V-B2), but ranks lower on Chat—suggesting potential bias in GPT-4-based evaluation, which may favor certain stylistic responses. See the supplementary material (Section V) for illustrative examples supporting this observation. Nevertheless, Falcon3-Audio-1B performs strongly, surpassing several larger models (7B–13B), highlighting the efficiency of our architecture and training strategy in open-ended audio understanding.

VI. CONCLUSION AND FUTURE DIRECTIONS

We presented Falcon3-Audio, a family of ALMs that demonstrate strong performance across diverse benchmarks through a simple, transparent, and data-efficient design built entirely on public resources. While several concurrent efforts have recently emerged, Falcon3-Audio remains among the most reproducible and data-efficient models to date. In future work, we aim to:

- Expand the training corpus to include more diverse and multilingual audio data.
- Explore larger models and reinforcement learning-based alignment techniques.
- Extend the framework to multimodal settings including audio, vision, and text.
- Investigate instruction-driven audio synthesis.

REFERENCES

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altmenschmidt, Sam Altman, Shyamal Anadkat, et al., “Gpt-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [2] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al., “Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context,” *arXiv preprint arXiv:2403.05530*, 2024.
- [3] Aaron Grattafiori et al., “The llama 3 herd of models,” 2024.
- [4] Falcon-LLM Team, “The falcon 3 family of open models,” December 2024.
- [5] An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al., “Qwen2 technical report,” *arXiv preprint arXiv:2407.10671*, 2024.
- [6] Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, et al., “Phi-3 technical report: A highly capable language model locally on your phone,” *arXiv preprint arXiv:2404.14219*, 2024.
- [7] Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing, “Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality,” March 2023.
- [8] Guan Wang, Sijie Cheng, Xianyu Zhan, Xiangang Li, Sen Song, and Yang Liu, “Openchat: Advancing open-source language models with mixed-quality data,” *arXiv preprint arXiv:2309.11235*, 2023.
- [9] Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen, “Dora: Weight-decomposed low-rank adaptation,” in *Forty-first International Conference on Machine Learning*, 2024.
- [10] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al., “Flamingo: a visual language model for few-shot learning,” *Advances in neural information processing systems*, vol. 35, pp. 23716–23736, 2022.
- [11] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee, “Improved baselines with visual instruction tuning,” 2024.
- [12] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Conghui He, Jiaqi Wang, Feng Zhao, and Dahua Lin, “Sharegpt4v: Improving large multi-modal models with better captions,” in *European Conference on Computer Vision*. Springer, 2025, pp. 370–387.
- [13] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou, “Qwen-vl: A frontier large vision-language model with versatile abilities,” *arXiv preprint arXiv:2308.12966*, 2023.
- [14] Hugo Laurençon, Lucile Saulnier, Léo Tronchon, Stas Bekman, Amanpreet Singh, Anton Lozhkov, Thomas Wang, Siddharth Karamcheti, Alexander M. Rush, Douwe Kiela, Matthieu Cord, and Victor Sanh, “Obelics: An open web-scale filtered dataset of interleaved image-text documents,” 2023.
- [15] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever, “Learning transferable visual models from natural language supervision,” 2021.
- [16] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer, “Sigmo loss for language image pre-training,” 2023.
- [17] Andrey Guzhov, Federico Raue, Jörn Hees, and Andreas Dengel, “Audioclip: Extending clip to image, text and audio,” in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 976–980.
- [18] Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang, “Clap learning audio concepts from natural language supervision,” in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [19] Shansong Liu, Atin Sakkeer Hussain, Chenshuo Sun, and Ying Shan, “Music understanding llama: Advancing text-to-music generation with question answering and captioning,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 286–290.
- [20] Zihao Deng, Yinghao Ma, Yudong Liu, Rongchen Guo, Ge Zhang, Wenhao Chen, Wenhao Huang, and Emmanouil Benetos, “Musilingo: Bridging music and text with pre-trained language models for music captioning and query response,” *arXiv preprint arXiv:2309.08730*, 2023.
- [21] Yuan Gong, Hongyin Luo, Alexander H. Liu, Leonid Karlinsky, and James Glass, “Listen, think, and understand,” 2024.
- [22] Ziyang Ma, Guanrou Yang, Yifan Yang, Zhifu Gao, Jiaming Wang, Zhihao Du, Fan Yu, Qian Chen, Siqi Zheng, Shiliang Zhang, et al., “An embarrassingly simple approach for llm with strong asr capacity,” *arXiv preprint arXiv:2402.08846*, 2024.
- [23] Shengpeng Ji, Ziyue Jiang, Wen Wang, Yifu Chen, Minghui Fang, Jialong Zuo, Qian Yang, Xize Cheng, Zehan Wang, Ruiqi Li, et al., “Wavtokenizer: an efficient acoustic discrete codec tokenizer for audio language modeling,” *arXiv preprint arXiv:2408.16532*, 2024.
- [24] Soham Deshmukh, Benjamin Elizalde, Rita Singh, and Huaming Wang, “Pengi: An audio language model for audio tasks,” *Advances in Neural Information Processing Systems*, vol. 36, pp. 18090–18108, 2023.
- [25] Yuan Gong, Alexander H. Liu, Hongyin Luo, Leonid Karlinsky, and James Glass, “Joint audio and speech understanding,” in *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2023, pp. 1–8.
- [26] Changli Tang, Wenyi Yu, Guangzhi Sun, Xianzhao Chen, Tian Tan, Wei Li, Lu Lu, Zejun Ma, and Chao Zhang, “Salmonn: Towards generic hearing abilities for large language models,” 2024.
- [27] Sreyan Ghosh, Sonal Kumar, Ashish Seth, Chandra Kiran Reddy Evuru, Utkarsh Tyagi, S Sakshi, Oriol Nieto, Ramani Duraiswami, and Dinesh Manocha, “Gama: A large audio-language model with advanced audio understanding and complex reasoning abilities,” *arXiv preprint arXiv:2406.11768*, 2024.
- [28] Yunfei Chu, Jin Xu, Xiaohuan Zhou, Qian Yang, Shiliang Zhang, Zhijie Yan, Chang Zhou, and Jingren Zhou, “Qwen-audio: Advancing universal audio understanding via unified large-scale audio-language models,” 2023.
- [29] Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, Chang Zhou, and Jingren Zhou, “Qwen2-audio technical report,” 2024.
- [30] Sanyuan Chen, Yu Wu, Chengyi Wang, Shujie Liu, Daniel Tompkins, Zhuo Chen, and Furu Wei, “Beats: Audio pre-training with acoustic tokenizers,” 2022.
- [31] Gang Li, Jizhong Liu, Heinrich Dinkel, Yadong Niu, Junbo Zhang, and Jian Luan, “Reinforcement learning outperforms supervised fine-tuning: A case study on audio question answering,” *arXiv preprint arXiv:2503.11197*, 2025.
- [32] Sreyan Ghosh, Zhifeng Kong, Sonal Kumar, S Sakshi, Jaehyeon Kim, Wei Ping, Rafael Valle, Dinesh Manocha, and Bryan Catanzaro, “Audio flamingo 2: An audio-language model with long-audio understanding and expert reasoning abilities,” *arXiv preprint arXiv:2503.03983*, 2025.
- [33] Abdelrahman Abouelenin, Atabak Ashfaq, Adam Atkinson, Hany Awadalla, Nguyen Bach, Jianmin Bao, Alon Benhaim, Martin Cai, Vishrav Chaudhary, Congcong Chen, et al., “Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras,” *arXiv preprint arXiv:2503.01743*, 2025.
- [34] Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, et al., “Qwen2. 5-omni technical report,” *arXiv preprint arXiv:2503.20215*, 2025.
- [35] Qian Yang, Jin Xu, Wenrui Liu, Yunfei Chu, Ziyue Jiang, Xiaohuan Zhou, Yichong Leng, Yuanjun Lv, Zhou Zhao, Chang Zhou, and Jingren Zhou, “Air-bench: Benchmarking large audio-language models via generative comprehension,” 2024.
- [36] S Sakshi, Utkarsh Tyagi, Sonal Kumar, Ashish Seth, Rameswaran Selvakumar, Oriol Nieto, Ramani Duraiswami, Sreyan Ghosh, and Dinesh Manocha, “Mmau: A massive multi-task audio understanding and reasoning benchmark,” *arXiv preprint arXiv:2410.19168*, 2024.
- [37] Yuan Gong, Yu-An Chung, and James Glass, “Ast: Audio spectrogram transformer,” *arXiv preprint arXiv:2104.01778*, 2021.
- [38] Rohit Girdhar, Alaaeldin El-Nouby, Zhuang Liu, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra, “Imagebind: One embedding space to bind them all,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2023, pp. 15180–15190.
- [39] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever, “Robust speech recognition via large-scale weak supervision,” in *International conference on machine learning*. PMLR, 2023, pp. 28492–28518.

- [40] Yuan Gong, Sameer Khurana, Leonid Karlinsky, and James Glass, “Whisper-at: Noise-robust automatic speech recognizers are also strong general audio event taggers,” in *INTERSPEECH 2023*, Aug. 2023, ISCA.
- [41] Sarvam AI, “Shuka v1 - ai-powered conversations,” 2024, Accessed: November 29, 2024.
- [42] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee, “Improved baselines with visual instruction tuning,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 26296–26306.
- [43] Jort F Gemmeke, Daniel PW Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R Channing Moore, Manoj Plakal, and Marvin Ritter, “Audio set: An ontology and human-labeled dataset for audio events,” in *2017 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2017, pp. 776–780.
- [44] Chris Dongjoo Kim, Byeongchang Kim, Hyunmin Lee, and Gunhee Kim, “Audiocaps: Generating captions for audios in the wild,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 119–132.
- [45] Eduardo Fonseca, Xavier Favory, Jordi Pons, Frederic Font, and Xavier Serra, “Fsd50k: an open dataset of human-labeled sound events,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 829–852, 2021.
- [46] Eduardo Fonseca, Jordi Pons Puig, Xavier Favory, Frederic Font Corbera, Dmitry Bogdanov, Andres Ferraro, Sergio Oramas, Alastair Porter, and Xavier Serra, “Freesound datasets: a platform for the creation of open audio datasets,” in *Hu X, Cunningham SJ, Turnbull D, Duan Z, editors. Proceedings of the 18th ISMIR Conference; 2017 oct 23-27; Suzhou, China.[Canada]: International Society for Music Information Retrieval; 2017. p. 486-93*. International Society for Music Information Retrieval (ISMIR), 2017.
- [47] Heiga Zen, Viet Dang, Rob Clark, Yu Zhang, Ron J Weiss, Ye Jia, Zhifeng Chen, and Yonghui Wu, “Libritts: A corpus derived from librispeech for text-to-speech,” *arXiv preprint arXiv:1904.02882*, 2019.
- [48] Honglie Chen, Weidi Xie, Andrea Vedaldi, and Andrew Zisserman, “Vggsound: A large-scale audio-visual dataset,” in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 721–725.
- [49] Arsha Nagrani, Joon Son Chung, and Andrew Zisserman, “Voxceleb: a large-scale speaker identification dataset,” *arXiv preprint arXiv:1706.08612*, 2017.
- [50] JanAI, “Instruction speech,” June 2024.